Reward Shaping Is Noise Shaping

What moves the accuracy–diversity curve in RL for reasoning

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The problem

- RL with verifiable rewards: the model writes answers, a checker grades them, the update is $g = \sum_i A_i \nabla \log \pi(y_i)$.
- Many variants of the advantage A_i: reward right answers (PSR), penalize wrong ones (NSR), change the baseline, aim at pass@k.
- Chosen for **diversity** (pass@k at large k), which RL is known to reduce.
- **The flaw:** methods are compared at a fixed training step. A slower method is less sharpened, so it looks more diverse.
- The fair comparison: diversity *at the same accuracy*.

The idea in one slide

Same direction, different noise

Every correctness-only advantage pushes the model the same way on average; they differ in **noise**, and wrong answers carry most of it. Noise sets *how fast* a run moves along one accuracy–diversity curve, not the curve.

Diversity is spread across questions

pass@k at matched pass@1 depends only on how evenly success is spread over questions. To move the curve, a method must bring new signal to rarely solved questions.

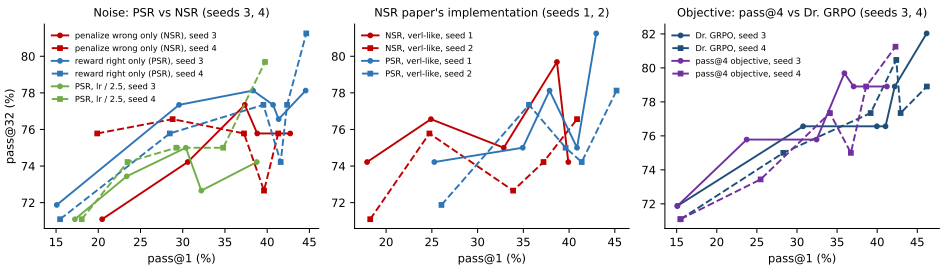
Theory: four numbers per question

For any rule that decides when to stop sampling and which advantages to give from the rewards:

$$\mathbb{E}g = c(p)\,\nabla p, \qquad \mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p)$$

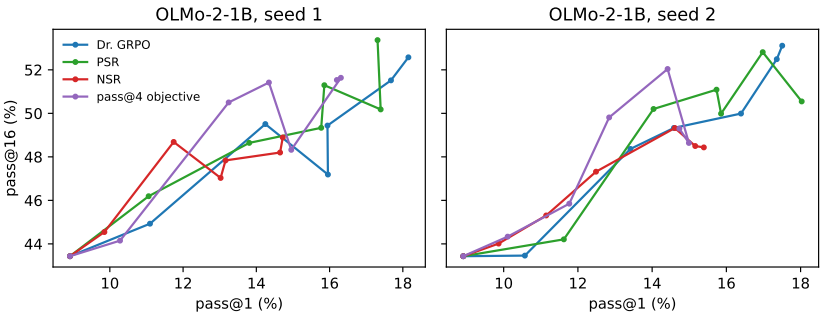
- F_1, F_0: mean squared gradient of a right / wrong answer.
- PSR and NSR: same mean $n\nabla p$ (classical baseline invariance); NSR = PSR + zero-mean noise.
- Correctness-only rules can only **reweight questions** and **change noise**.
- Measured: wrong answers carry **2.1–3.4**$\times$ larger squared gradients (both families); fades to 1.5$\times$ in training.

Diagnosis 1: NSR is a slower PSR (Qwen2.5-Math-1.5B)



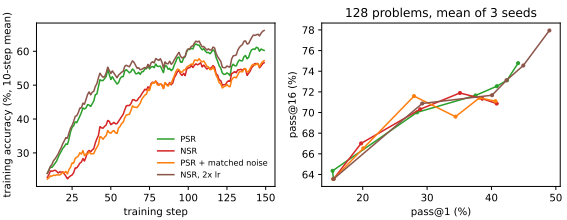
NSR, the NSR paper's own implementation, and a pass@4 objective: all on one curve.

Diagnosis 2: replicated on a second family (OLMo-2-1B)



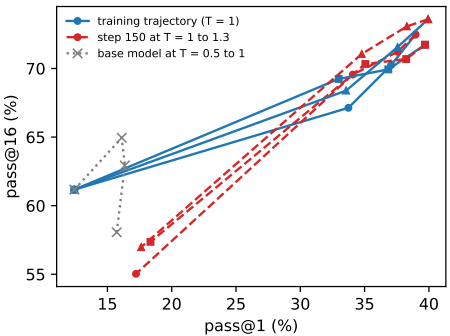
NSR vs PSR at matched accuracy: -0.30, 90% $[-1.28, +0.77]$: **equivalent within** ± 2. NSR 19–22$\times$ noisier.

Diagnosis 3: the noise is the cause



- Add random-sign noise of NSR's size to PSR (same mean).
- **PSR + noise is as slow as NSR**: $\phi = -0.11$ $[-0.22, -0.05]$ (NSR $= 0$, PSR $= 1$).
- NSR at $2\times$ learning rate recovers the lag.
- All on one curve (equivalent within ± 2, 3 fresh seeds).
- Adam: noise does not shrink the steps, it misdirects them.

Diagnosis 4: not a temperature effect



- On fixed text, entropy barely changes (1.081 $\rightarrow$ 1.064); on own samples it falls 3–4$\times$: RL steers into confident paths.
- A late checkpoint at a raised temperature does **not** beat the training curve ($D = -1.42$ on 511 problems; 0.0 at the entropy-matched temperature).
- RL is far above anything temperature does for the base model: RL adds ability.

What the diagnosis says a method must do

Finding	Requirement
Correctness-only rules share one curve	bring information the samples lack
Diversity = spread of success across questions	raise rarely solved questions
Rarely solved: groups of 8 are all wrong	supply signal there
No global sharpening; entropy control blocks learning	do not regulate entropy
Noise slows learning	keep Dr. GRPO as the base

Method: coverage-weighted success replay

Store verified successes; each step replay 8 questions, sampled $\propto k\,p(1-p)^{k-1} = \partial\,\mathrm{pass@}k/\partial \log p$: the questions whose pass@k can still rise.

Pilot (1 seed, validation): memory is the only family that lifts

Arm	pass@1	pass@16	gain at matched accuracy
Coverage memory $\lambda = 1$	35.8	73.3	$+2.1$
Uniform memory $\lambda = 1$	38.6	72.4	$+0.2$
Uniform memory $\lambda = 4$	35.3	74.2	$+3.2$ (too slow; excluded)
Entropy bonus	38.9	72.7	≈ 0
Unlikeliness reward	38.7	71.7	≈ -0.5
Adaptive entropy	25.5	65.0	-2.5
Early-token entropy floor	12.8	56.1	collapsed

Main runs: interim (3 fresh seeds, 511 test problems)

- **Training done** (12 runs): all four arms train equally (training accuracy 60–63, greedy MATH-500 43–47).
- **Evaluation in progress** (seed 10 so far):

Seed 10	pass@1	pass@16	vs Dr. GRPO (incl. best temperature)
Dr. GRPO, step 50 / 100 / 150	37.3 / 40.6 / 44.6	72.4 / 71.9 / 75.2	reference; greedy test 48.2%
Coverage memory, step 50	36.4	71.8	−0.2
Coverage memory, step 100	39.6	72.9	+0.3

- Interim: **no lift yet in this seed**; the verdict needs all 3 seeds (9 points).

Where we stand

Stage	Independent review (Codex)	Why
Qwen only	4/10	claims too broad, cause not shown
+ noise is the cause	5/10	causal test passed
+ OLMo replication	**6/10 (weak accept)**	same curve on a second family
Target	7/10	a method that moves the curve, replicated

- **If memory moves the curve:** OLMo replication, then the full ICML paper.
- **If not:** a strong diagnostic paper, with an audit showing that the popular diversity fixes (NSR, entropy control, unlikeliness, pass@k weights) do not move the curve, and why.
- ICML 2027 (deadline usually late January); draft in `paper/icml/`.